NVIDIA SDK + FBOSS: Building Production-Ready AI Network Fabrics

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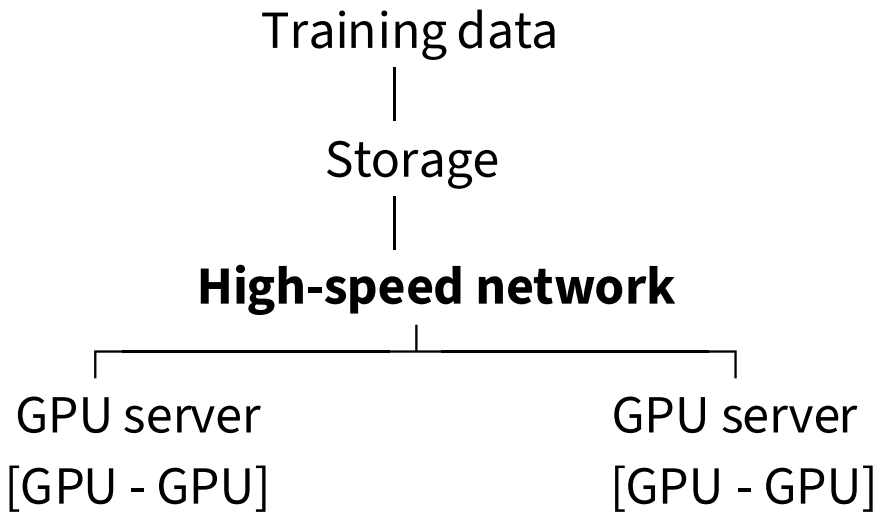
Networking Team for large-scale AI / GPU clusters

Agenda

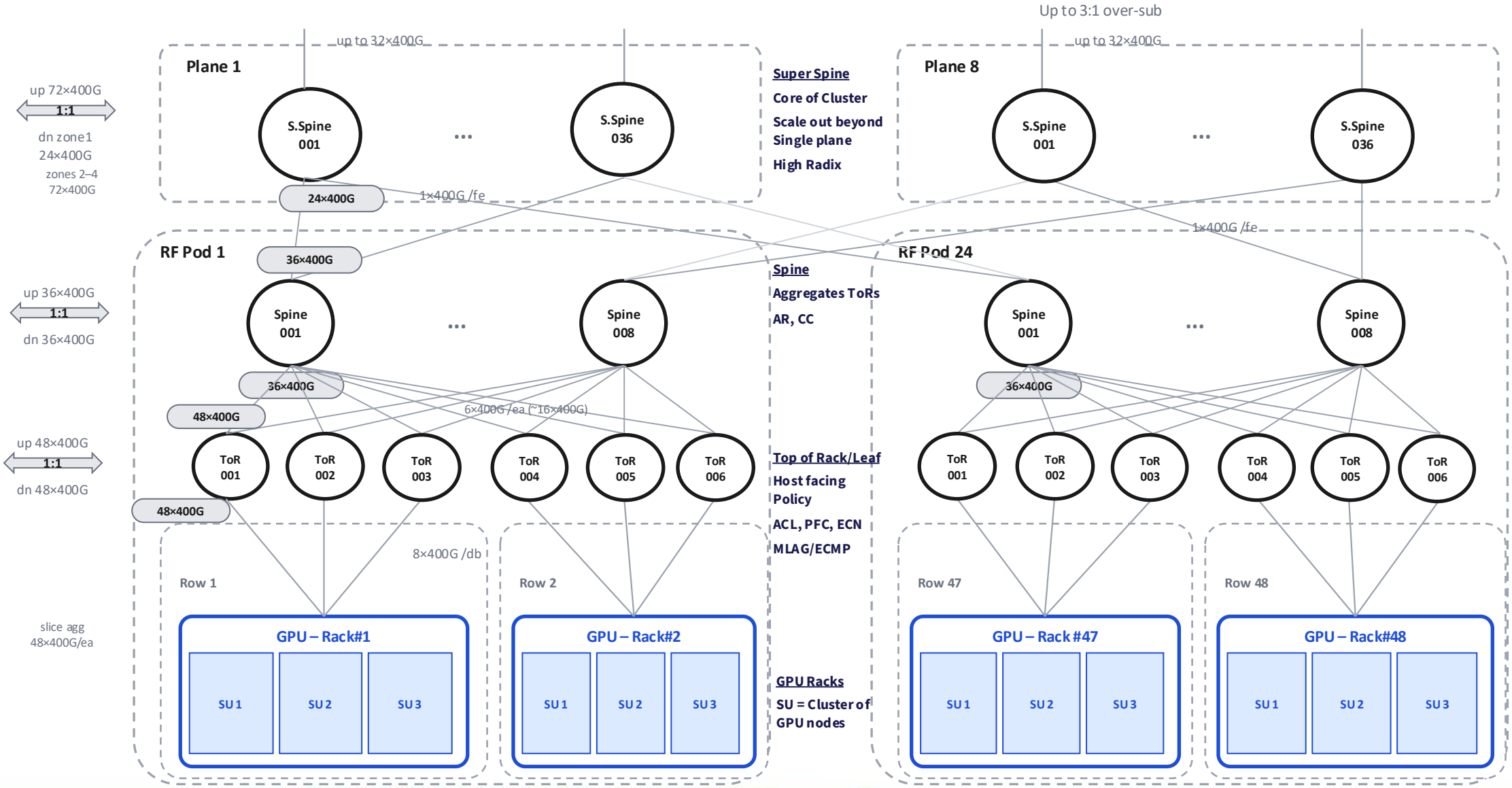
- **Part 1 — AI fabrics and their challenges**
  - AI networks: Characteristics
  - Topology of an AI cluster
- **Part 2 — The building blocks**
  - NVIDIA Spectrum-X Switch
  - NVIDIA Software: SDK & SAI
  - NVIDIA Simulator
  - FBOSS — Meta's network operating system
- **Part 3 — Bringing it all together**
  - CI/CD & Test-driven development
  - Mutual Evolution / Joint collaboration
  - Rapid Integration & Timelines
- **Summary & Call-to-Action**

AI Networks: Characteristics

- **What are AI networks?**
  - Physical connections between GPUs, servers, and storage in an AI data center
  - Purpose-built fabrics connecting thousands of GPUs for Training & Inference
  - East-West heavy traffic: Support Collective operations (All-Reduce, All-Gather, All-to-All etc.)
- **Complexity and challenges**
  - Require ultra-high bandwidth, ultra-low latency, and predictable tail latency
  - Huge scale means a huge fault blast radius
  - Need AI networking features – Adaptive Routing and Congestion Control
  - AI buildout pace requires solutions to be delivered quickly



Typical AI Cluster Scale



One zone: 288 super spines · 192 spines · 144 ToRs · 48 racks · 3,456 GPUs — every layer must be programmed, tested and upgraded without stalling the fleet.

AI network building blocks





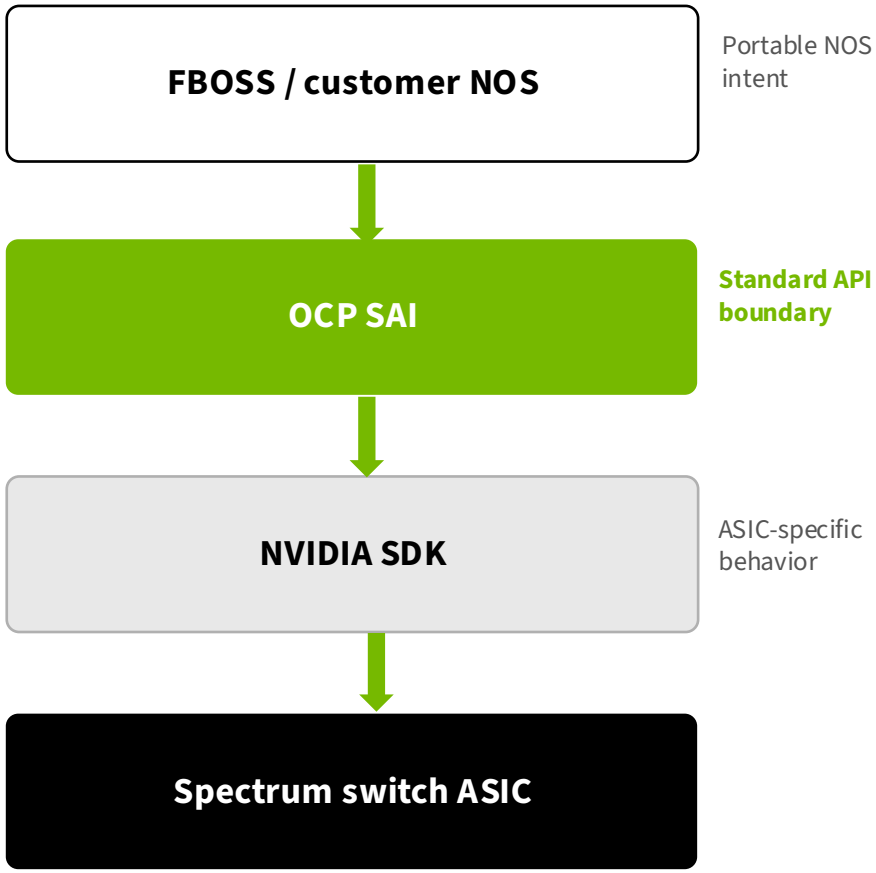
NVIDIA Spectrum-6 100T Switch

- **Features**
  - 102.4 Tb/s switch with 512 x 200G or 128 x 800G ports
  - Pluggable and Co-Packaged Optics
  - Adaptive Routing, Congestion Control, Jitter-free communications
- **Telemetry**
  - Fabric Visibility: What just Happened (WJH), In-band Telemetry
  - Deep visibility into pipelines and queues; counters & histograms

NVIDIA Software: SDK & OCP-Compliant SAI

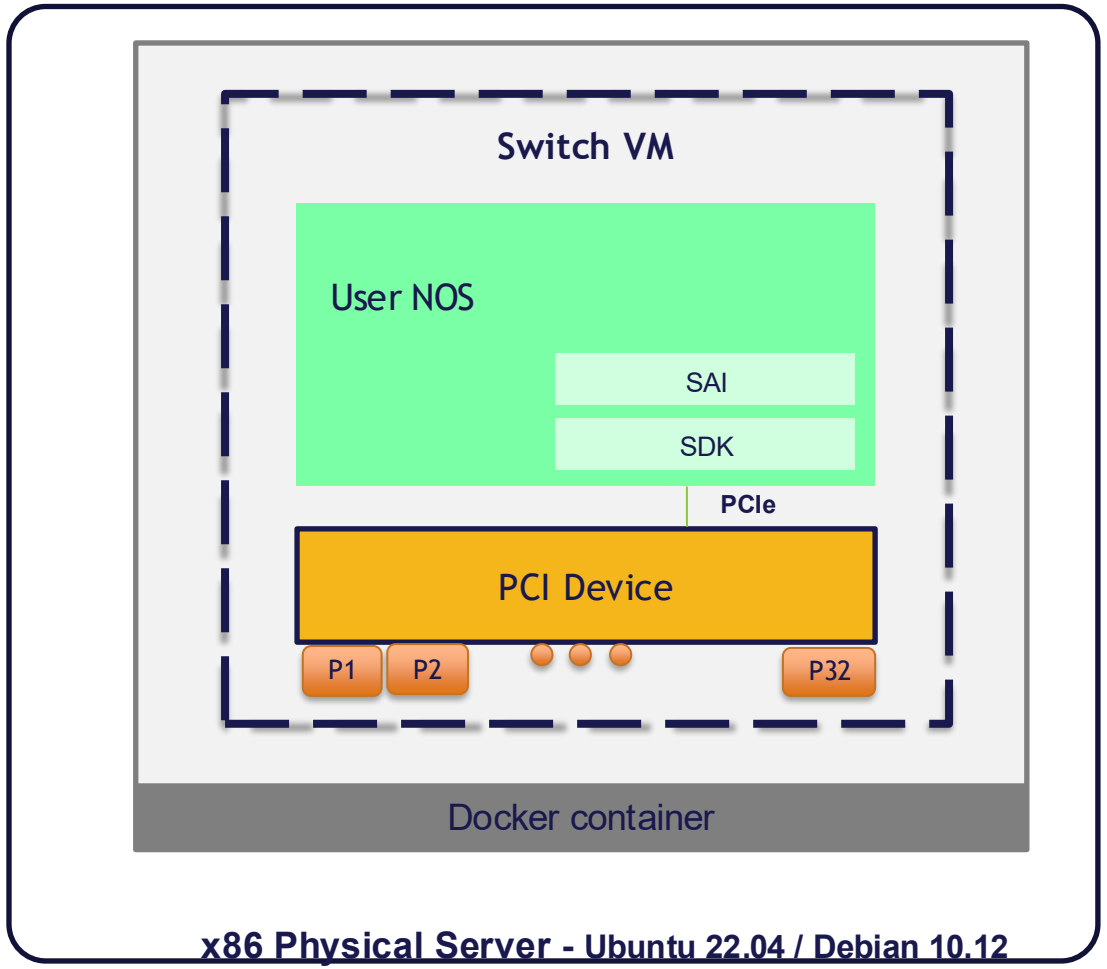
- **NVIDIA SDK**
  - APIs & services to program and manage the ASICs
  - Path: silicon capabilities → deployable network features
  - Diagnostics & CLI for easier debug
  - Single Spectrum SDK across all chip generations

- **NVIDIA SAI (OCP compliant)**
  - SAI provides common APIs/constructs across vendors
  - Decouples NOS logic from silicon-specific details
  - Enables faster platform bring-up & new vendor onboarding



NVIDIA Functional Simulator for Spectrum Switch

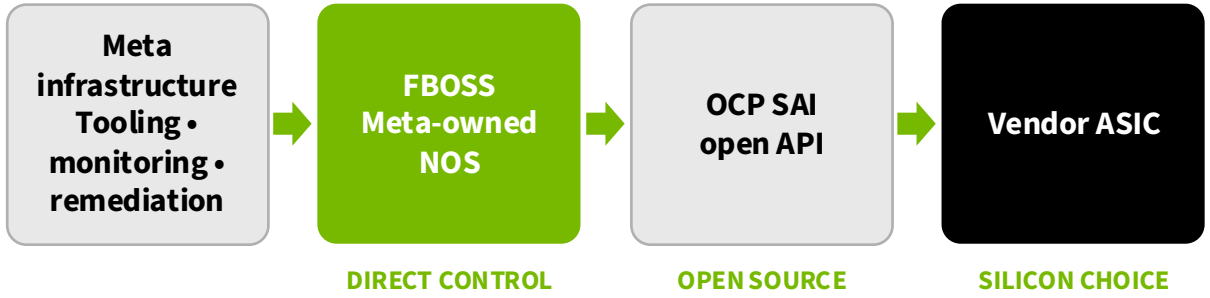
- **NVIDIA SimX**
  - Software modeled ASIC over PCI inside a VM
  - NOS software talks to it like real hardware
  - Allows to develop & test SDK stack
  - Software no longer gated solely on HW arrival
  - Supports forwarding tables, L2, L3, ACL..
  - Does not simulate QoS, L1 protocols, etc.

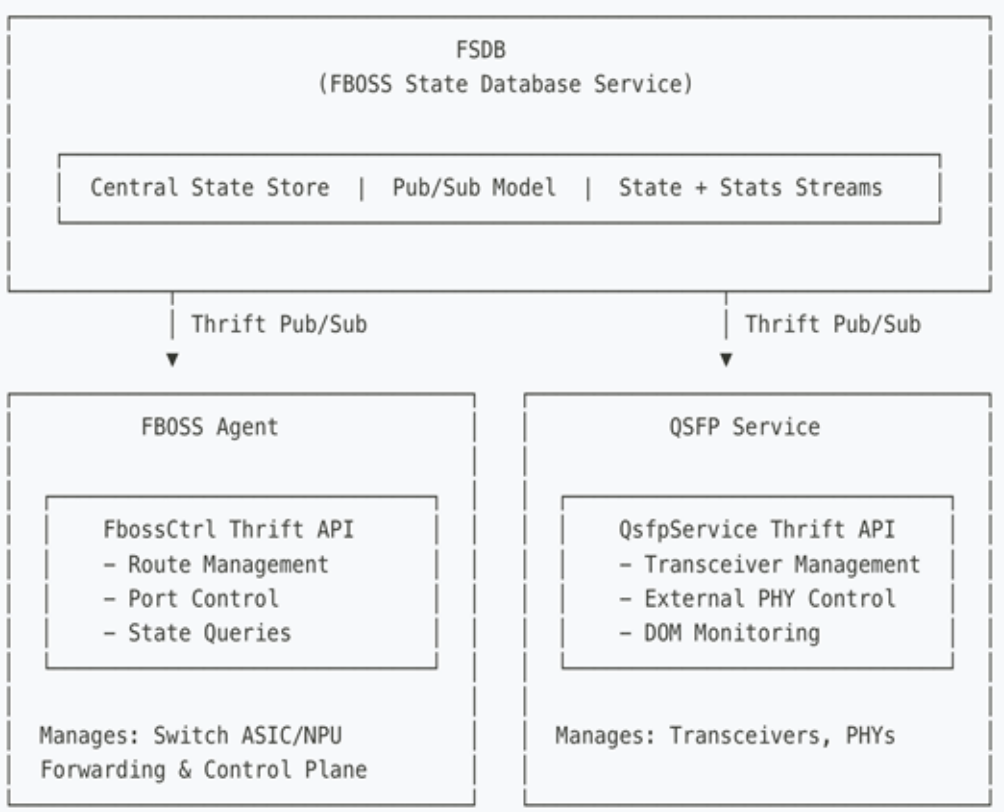


NOS and ASIC work proceed in parallel; hardware availability is not a bottleneck

Meta's Network Operating System: FBOSS

- Open source NOS; developed over SAI
- Meta owns and controls the NOS & its features
- Infra-wide tooling is closely coupled with FBOSS
- CI/CD Model for NOS development
  - Unit / agent / feature / HW-facing tests gate merges
  - Earlier detection of SDK / SAI / ASIC mismatches





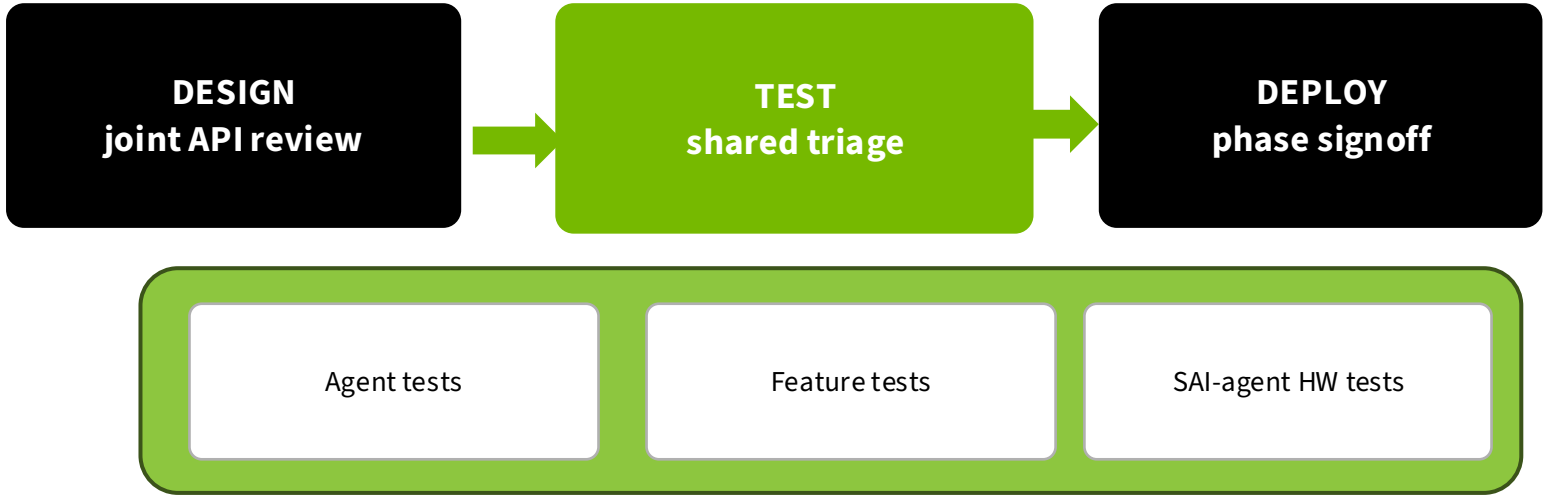
Production-proven NOS for AI fabrics: Scale & Reliability

Bringing it all together



Test-Driven Development: NVIDIA ↔ Meta at Every Phase

- Collaboration through Design, Test & Deployment; Joint triage of behavioral gaps
- Features land with tests as the contract (Agent, Feature, HW test suites evolve continuously)
- Simulator + HW pipelines validate early; regression suite grows with each milestone
- Tests run and validated at NVIDIA before delivery
- Early feature adoption on both sides → never a dead-on-arrival scenario



Earlier feature adoption • Faster convergence • No dead-on-arrival

Faster regression convergence and predictable onboarding cadence

Mutual Evolution / Joint Collaboration

Enhanced NVIDIA-SDK & Meta-FBOSS ; some examples..

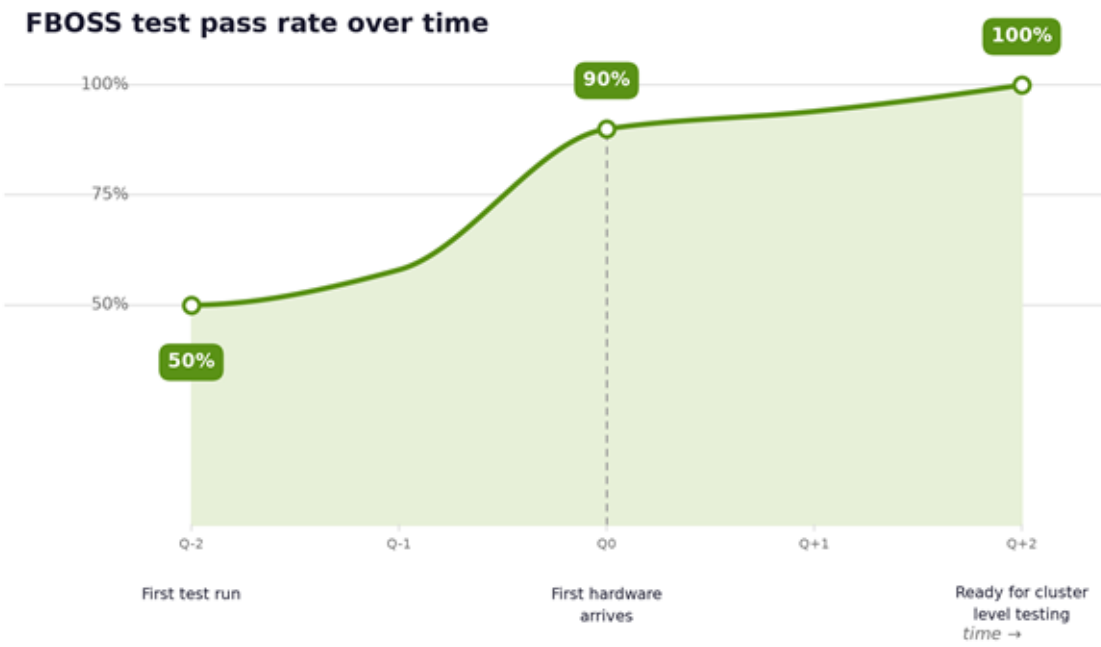
- Enhanced SDK-ISSU implementation
  - Continuity across upgrades / warm upgrade without impacting traffic
  - Full resource usage; export state to file & reload

- Debuggability in Network
  - Hardware drop reasons & Counters
  - Mirror on Drop (MoD) utilizing Telemetry Aggregation Cache (TAC)
  - Added SDK & Device Health-check feature

- Cable Length estimation
  - Added support using PTP

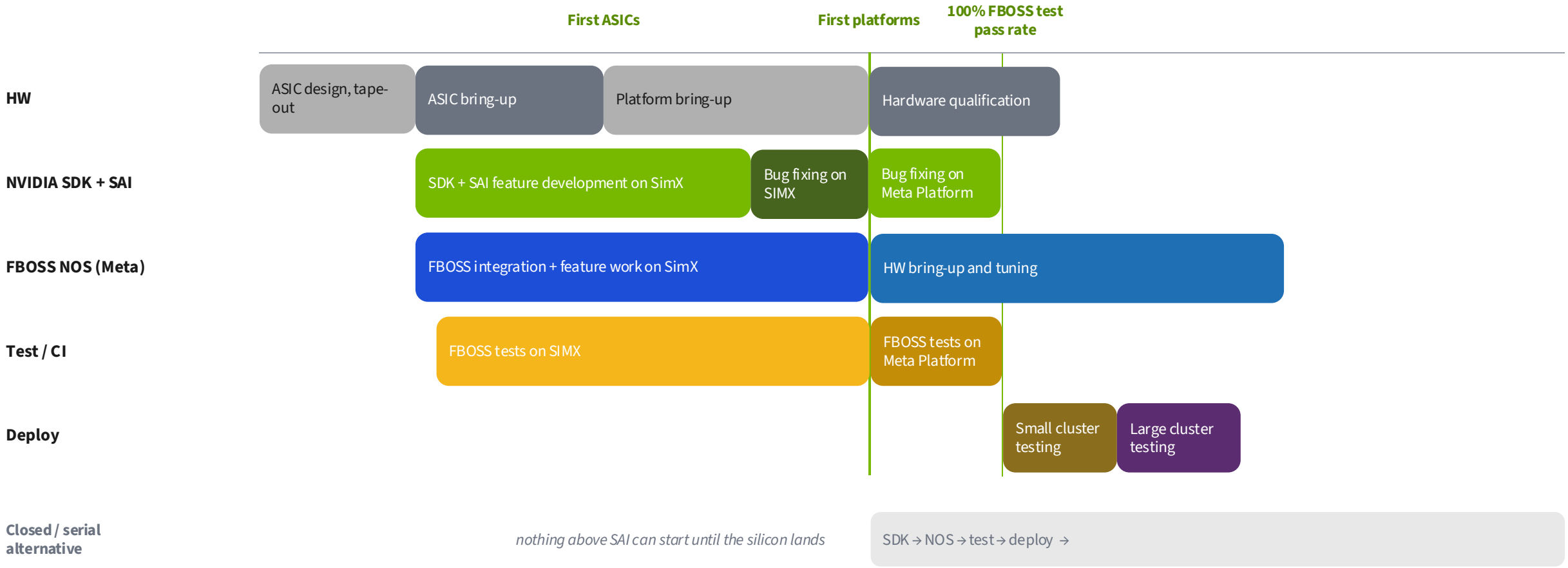
SAI + OSS FBOSS + SIMX = rapid integration

- Develop needed features in FBOSS / Nvidia SAI & execute tests on SIMX / NV alternate platform **in ~6 weeks**
- 90% pass rate within **~1 week** after Meta's Hardware first arrives
- Quickly resolve platform specific issues -> full pass rate and ready for cluster qualification in **~2 quarters**



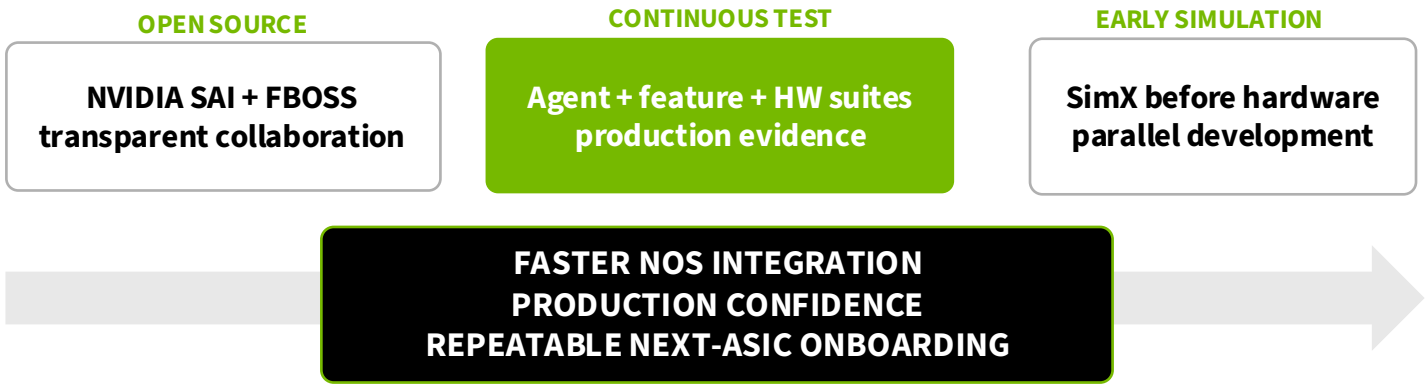
Parallel by Construction: Open SAI + SimX Compress the Schedule

- SAI contract decouples the schedule: SDK, NOS, test and deployment tracks all start before silicon exists
- So hardware arrival is a qualification milestone, not the starting gun



Summary: Open Interfaces → Faster AI Fabric Onboarding

- NVIDIA SAI (OCP) + open-source FBOSS reduced integration friction and accelerated NOS onboarding
- Spectrum SimX removed the hardware bottleneck for next-gen bring-up
- CI/CD + Test-driven partnership across design / test / deploy prevents dead-on-arrival outcomes



Call to Action

- Share your own experiences in accelerating NOS development
- Share your own experiences in SDK/SAI integration in Emulated environments


Thank You!
Questions? | NVIDIA SDK + FBOSS | Building Production-Ready AI Network Fabrics
OCP GLOBAL SUMMIT
October 12-15, 2026
San Jose, California